Chapter 6 exercises. State meanings and complete proofs

DM06-E01 • Empty words and countability

Compare Σ^*, Σ^+, and $\varnothing$, $\{\varepsilon\}$. Prove finite words over a finite nonempty alphabet countably infinite. Must a general language's L^+ exclude the empty word? Give an exact criterion.

DM06-E02 • A complete length-modulo-three DFA

Over $\{a, b\}$, design a total DFA accepting lengths divisible by three, including empty input. Give its five-tuple, complete transition table, and the path for abaabb.

DM06-E03 • A state proof for every input

For question two, induct on length to prove the final state $q_{|w| \bmod 3}$. If only the initial state changes to q_1, determine the new language by proof rather than guessing from short words.

DM06-E04 • Modulo three and final bit one

Design a DFA accepting binary values divisible by three and ending in one, permitting leading zeros but rejecting empty input. Give all product states, the full table and accepting set, trace 11011, and prove both state components' meanings.

DM06-E05 • Complementing partial transitions

A sole initial accepting state s on $\{a\}$ has no transitions; missing edges reject. Find its language and explain why switching its accepting mark fails to complement it. Complete the machine and construct the correct complement.

DM06-E06 • Accepting states need not reach each other

Over $\Sigma = \{a, b\}$, construct a total three-state DFA with two mutually unreachable accepting states that accepts every nonempty word but not the empty word. Give a complete table and general correctness proof. Are the accepting states equivalent?

DM06-E07 • Incorrect language identities

Give finite languages with $|AB| < |A||B|$. Refute $A(B \cap C) = AB \cap AC$ and prove the valid inclusion. By factor decomposition prove $(A \cup B)^* = (A^*B^*)^*$, including empty-language cases.

DM06-E08 • Carry states and padding

Use the serial adder for $9 = (1001)_2$ plus $7 = (0111)_2$, low bit first, followed by 00 padding. Give each pair, old carry, output, and new carry. Prove the k-bit invariant and explain how the final state recovers the answer without padding.

DM06-E09 • Overlapping matches and phase

Trace every state and output of the 101 Mealy machine on 1010101. Restrict reporting to input positions divisible by three, giving the nine-state product update and new output. Why does resetting to the initial state after a match lose information?

DM06-E10 • A delay-machine lower bound

Construct the two-tick delay Mealy machine with initial zero padding. Give its four-state transition/output table and trace 1101. Use distinguishable histories to prove four states necessary. Explain output length and flushing.

DM06-E11 • Determinizing an epsilon-NFA and a false complement

Consider the ε-NFA over $\{0, 1\}$: start state s has an ε-edge to p; p loops on both 0 and 1 and also has a 1-edge to q; q has a 0-edge to r. The sole accepting state is r, and unlisted edges do not exist. Find all reachable subsets and the complete DFA table. Trace 01100 and prove the subset invariant generally. Does swapping the NFA accepting states complement its language? Prove it or refute with a word.

DM06-E12 • Paths, elimination, and a regular expression

An NFA has start and sole accepting state p, another state q, a zero-loop at p, a one-edge p to q, and a one-edge q to p, with no other edges. Add fresh start and end states, eliminate q then p, and derive a regular expression with a path-semantics proof. Decide 1110 and 11011.

DM06-E13 • An impossibility proof must cover every split

Prove $\{0^n 1^{2n} : n \geq 0\}$ nonregular. Use pumping, identifying y in every valid split and the chosen repetition count. Independently use same-state prefixes and one shared suffix.

DM06-E14 • Shortest transfers and distinguishing words

In the lecture's S,A,B,T table, is the shortest S-to-T word unique? Prove the answer with BFS layers and the absence of repeated states on a shortest path.

Compare the two-state even-ones and even-zeros DFAs. List reachable product-state BFS layers and a shortest distinguishing word. Prove unequal general DFAs have a counterexample of length at most $|Q_1||Q_2| - 1$, including empty-input distinction as length zero.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 6.

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https://github.com/xinhi6716/computer-science-self-study

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